

Luigui Gallardo-Becerra

SOFTWARE ENGINEER & DATA SCIENTIST

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Software Engineer with 6+ years of experience in designing scalable data systems, automating complex workflows, and building cloud-native solutions. Proficient in Python, SQL, C#, with a strong background in backend development.

Experience

Graduate Research Assistant (Data Science & Bioinformatics)

Institute of Biotechnology, UNAM

January 2019 - July 2025

- Built an internal AMP research data portal using ASP.NET Core and clean architecture, with SQL Server and Dapper for querying and visualizing large-scale peptide prediction datasets.
- Engineered automated, containerized pipelines using Nextflow, Docker, and Snakemake for reproducible data processing.
- Optimized distributed computing systems and high-performance computing (HPC) workflows for large-scale analysis.
- Developed machine learning models to analyze complex datasets, resulting in 14 peer-reviewed publications.
- Conducted statistical analyses and feature engineering to identify trends and patterns in biological data.

Data Engineer

Appen

October 2016 - December 2018

- Built robust SQL-based ETL architectures for data integration and automated reporting systems.
- Developed and automated data collection workflows using Python and Bash.
- Preprocessed and analyzed large-scale datasets for machine learning projects.

Software Engineer (Internship)

Dept. of Computer Science, CUCEI

January 2016 - July 2016

- Developed a cloud-deployed web application using ASP.NET Core for patient record management, contributing to both backend and frontend development.

Education

National Autonomous University of Mexico (UNAM)

PH.D. IN COMPUTATIONAL BIOLOGY

Mexico City, Mexico

January 2019 - July 2025

National Autonomous University of Mexico (UNAM)

MASTER OF SCIENCE IN COMPUTATIONAL BIOLOGY

Mexico City, Mexico

August 2016 - January 2019

University of Guadalajara (UDG)

BACHELOR OF SCIENCE IN MOLECULAR BIOLOGY

Guadalajara, Mexico

August 2012 - January 2016

Key Skills

- **Programming & Frameworks:** C#, ASP.NET Core, Python, SQL, Bash, JavaScript, React, R
- **Cloud & Infrastructure:** AWS, Docker, CI/CD, Linux, Database Management, HPC
- **Tools:** Git, Jupyter, Snakemake, Nextflow
- **Data Engineering:** ETL Pipelines, Apache Spark, System Automation, Large-Scale Data Processing
- **Soft Skills:** Problem-solving, Technical Leadership, Project Management, Agile Collaboration
- **Languages:** English (Full professional proficiency), Spanish (native)